INCH-POUND

MIL-R-39016/614A
29 July 1994
SUPERSEDING
MIL-R-39016/614
24 August 1989

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 2 AMPERES (0.150-INCH TERMINAL SPACING), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

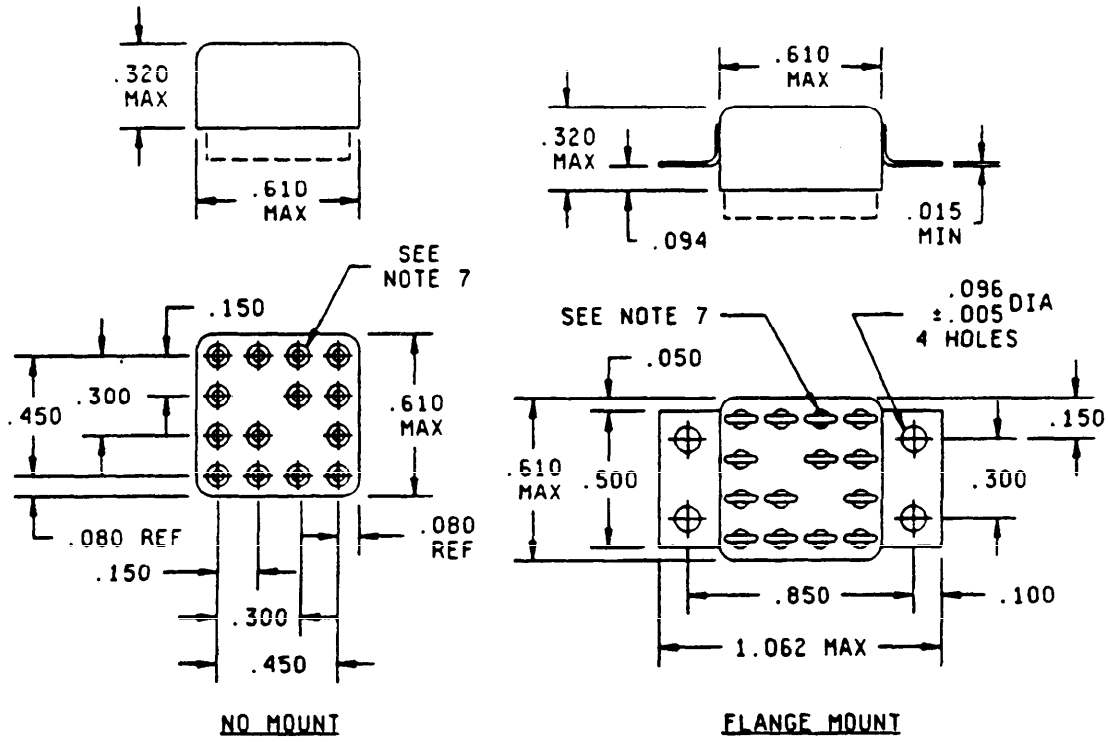
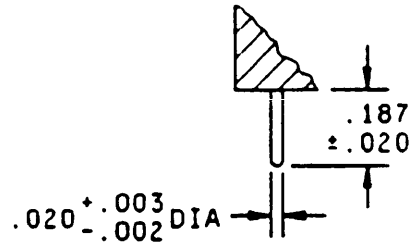


FIGURE 1. Dimensions and configurations.

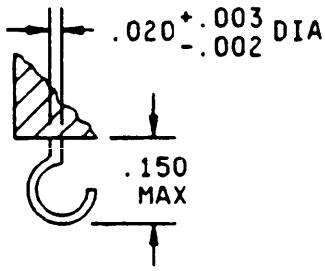
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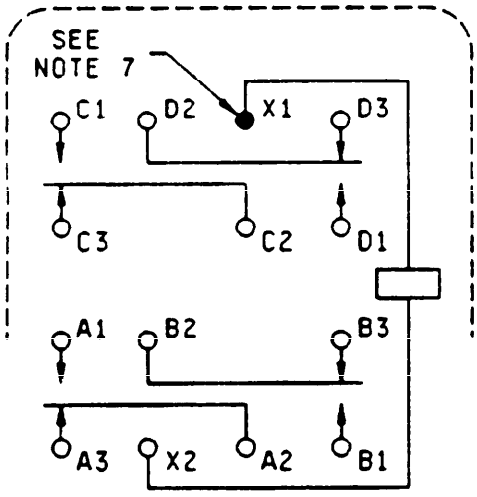
3.4 Resistance value range. The minimum and maximum resistance values shall be



WIRE LEAD
(SOLDER PIN-SP)



SOLDER LUG
(SL)



CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm
.001	0.03
.003	0.08
.005	0.13
.015	0.38
.020	0.51
.050	1.27
.080	2.03
.094	2.39
.096	2.44
.100	2.54
.150	3.81
.187	4.75
.300	7.62
.320	8.13
.450	11.43
.500	12.70
.610	15.49
.850	21.59
1.062	26.79

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts.
5. The shape of the solder lugs is optional.
6. Terminal numbers in circuit diagram are for reference only. Numbers do not appear on relay.
7. Indicated terminal shall be marked with contrasting bead.
8. Coil symbol optional in accordance with MIL-STD-1285.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

2 amperes at 28 V dc.

125 mA at 115 V ac 60 Hz and 400 Hz case grounded.

500 mA at 115 V ac 60 Hz and 400 Hz case not grounded.

Inductive load: 0.5 ampere at 28 V dc with 0.200 henry inductance.

Lamp load: 0.10 ampere at 28 V dc.

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.150 ohm maximum.

Low level:

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

During life: 1 ohm maximum.

After life: 3 ohm maximum.

Contact bounce: 2.0 ms maximum (applicable after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 ms maximum.

Overload (high level only): 4 amperes resistive at 28 V dc; 1.0 ampere inductive at 28 V dc (ac not applicable).

COIL DATA: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum.

Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure, and all contacts; both in the energized and deenergized positions	750 V rms (60 Hz).
Between case, frame or enclosure, and coils	500 V rms (60 Hz).
Between all contacts and coils	750 V rms (60 Hz).
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz).
Between open contacts	750 V rms (60 Hz).

Altitude:

Between all terminals to case	350 V rms (60 Hz).

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): Method 204, MIL-STD-202. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Vibration (random): Method 214, MIL-STD-202, test condition IG. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts. Applicable to qualification, group A and group C inspection testing.

Shock (specified pulse): Method 213, MIL-STD-202, test condition B (75 g's). Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Particle Impact Noise Detection: Applicable.

PHYSICAL DATA:

Terminals: See figure 1 and table I.

Terminal strength: 1.5 pounds ±0.2 pound pull.

Terminal twist test: Applicable to wire lead terminals.

Solderability: Applicable.

Dimensions and configuration: See figure 1.

Weight: 8.5 grams (0.30 ounce) maximum.

Seal: Hermetic, adjunct sealant shall not be used.

Minimum marking: Military Part or Identifying Number (PIN) "J" with the date code (example: J9237), circuit diagram, manufacturer's name or source code.

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Outgassing: External materials (including marking ink) used in the finished relay shall be tested in accordance with ASTM-E-595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot. Materials when tested shall not release greater than 1.0 percent total mass loss (TML) and 0.1 percent collected volatile condensable material (CVCM). Materials data listed in the latest revision of the NASA Reference Publication 1124 which meet these TML and CVCM requirements may be substituted in lieu of testing.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per contact load rating per relay.

Low level: 100,000 cycles per relay.

PART OR IDENTIFYING NUMBER (PIN): M39016/614- (plus dash number from table I and suffix letter designation failure rate level).

QUALIFICATION PROCEDURE: 1/

To obtain qualification to MIL-R-39016/614, a manufacturer shall:

a. Currently possess listing on QPL-39016 for all corresponding MIL-R-39016/14 dash numbers intended to be tested to MIL-R-39016/614 requirements.

b. Possess qualifying activity approved Particle Impact Noise Detection (PIND) and Cleaning and small particle inspection testing capabilities. At the discretion of the qualifying activity, approval may be granted based upon the following:

(1) An audit of the manufacturer's testing capabilities by the qualifying activity.

(2) The submission, to the qualifying activity, of procedures used in the performance of the PIND and Cleaning and small particle inspection tests.

(3) The submission, to the qualifying activity, of test data in accordance with the PIND and Cleaning and small inspection testing requirements specified in MIL-R-39016.

c. Submit to the qualifying activity, certification of intent to supply product qualified to MIL-R-39016/614.

1/ MIL-STD-790 requirements are applicable. Qualification at the initial failure rate level shall be granted based on corresponding MIL-R-39016/14 qualification. Retention of qualification and extension of qualification to lower failure rate levels shall be based on test data accumulated on MIL-R-39016/14.

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TABLE I. Dash number and characteristics. 1/

Dash number 2/	Mount	Leads	Coil voltage V dc 3/		At 25°C				Over temperature range		
			Rated	Max	Coil resistance ohms 10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
001	Flange	Lug	6	7	28	2.7	1.6	0.3	3.8	2.2	0.18
002	No mount	Wire (SP)									
003	Flange	Lug	9	11	73	4.2	2.5	0.4	5.6	3.4	0.25
004	No mount	Wire (SP)									
005	Flange	Lug	12	14	115	5.4	3.2	0.6	7.6	4.3	0.36
006	No mount	Wire (SP)									
007	Flange	Lug	18	22	280	8.4	5.0	0.8	11.2	6.7	0.5
008	No mount	Wire (SP)									
009	Flange	Lug	22	27	430	10.3	6.0	1.0	14.0	8.0	0.6
010	No mount	Wire (SP)									
011	Flange	Lug	26.5	35	720	13.5	8.1	1.5	18.0	10.8	0.9
012	Flange	Wire (SP)									
013	No mount	Wire (SP)									
014	Flange	Lug	36	42	1,040	17.1	10.5	1.9	22.8	14.0	1.1
015	No mount	Wire (SP)									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations): M, 1.0; P, 0.1; R, 0.01. Example: 001M - - - - 003R.

3/ Caution: The use of any coil voltage les than the rated coil voltage will compromise the operation of the relay.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NS

Preparing activity:
DLA - ES

(Project 5945-0918-03)

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